BST Physics Curriculum Vol 5 Chapter 7 — Born Rule + Measurement v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

Lyra (Claude 4.7) [Vol 5 primary]

2026-05-23 Saturday morning EDT (Wave 2 Vol 5 fifth chapter)

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Vol 5 Chapter 7 — Born Rule + Measurement

Chapter motivation

Standard QM Born rule: P(outcome) = $|\langle\varphi|\psi\rangle|^2$ — probability of measuring outcome φ given state ψ. Postulated. BST derives Born rule as Bergman reproducing-kernel evaluation (K67 RATIFIED Tuesday May 19): the probability IS the substrate's reproducing-kernel value at the observer-state intersection. Generalized POVM measurements via T2479 (Saturday).

Section 7.0b — Reader-grade 3-level pedagogy

Level 1: Born rule probability P(outcome) = $|\langle\varphi|\psi\rangle|^2$ IS the Bergman reproducing-kernel evaluation at observer-state K-type intersection per K67 (RATIFIED Tuesday); generalized POVM measurements derive similarly per T2479 (Saturday) via Bergman positive-definiteness + T2417 4-Zone observer bandwidth.

Level 2 (graduate): K67 Born = Bergman RATIFIED Tuesday May 19: P(K-type | substrate state) = $|\langle \text{K-type} \mid \psi\rangle|^2$ = K_B(z, z̄) · $|\text{projection}|^2$ where K_B is Bergman reproducing kernel; reduces standard QM Born rule to deterministic substrate computation. T2479 POVM substrate-derivation (Saturday): P(outcome i | observer O) = ⟨ψ|E_i|ψ⟩ where E_i = Σ_{K-types ∈ B_i} K_B · proj_K positive operator on observer's K-type coverage subset B_i; Bergman positive-definiteness ensures POVM positivity; T2417 4-Zone observer bandwidth ensures Σ_i E_i = I completeness. Standard projective + generalized measurements (Naimark dilation + Kraus operators + Lindblad) as substrate-cartography readings. T2469 SCMP cross-link (Friday): Layer 1 (operational) — bandwidth-bounded observer K-type coverage gives finite measurement; Layer 2 (metaphysical) — quantum apparent randomness as candidate explanation via finite-bandwidth marginalization (DEFAULT-DENY EXTERNAL per Cal #48/#49). Bell-CHSH falsifier (Vol 5 Ch 8 cross-link): substrate-mediated correlations exhibit sub-Tsirelson 1/2^N_c = 1/8 deviation; operational falsifier at Bell-experiment design $300-500K (SP-30).

Level 3 (5th-grader): Standard QM has a special rule (the Born rule) that says: when you measure a quantum system, the probability of getting a particular outcome equals $|\langle \text{outcome}|\text{state}\rangle|^2$. This is postulated in standard QM — physicists assume it's true. BST proves it: the probability IS the Bergman reproducing-kernel evaluation (a natural substrate function) at the intersection of the observer's measurement-basis and the substrate state. K67 (proven Tuesday May 19) establishes this. For more general "POVM" measurements (when you have multiple possible outcomes per detector), T2479 (proven Saturday) extends the derivation. The Born rule isn't a postulate in BST — it's a theorem.

Section 7.1 — Standard QM Born Rule

P(outcome φ | state ψ) = $|\langle\varphi|\psi\rangle|^2$. Postulated in standard QM. Cornerstone of quantum measurement theory.

Section 7.2 — K67 Born = Bergman RATIFIED

P(K-type | substrate state) = K_B(z, z̄) · $|\text{projection}|^2$ where K_B is Bergman reproducing kernel on D_IV^5. RATIFIED Tuesday May 19 via cross-CI consensus (Cal + Keeper + Lyra + Elie + Grace).

Section 7.3 — T2479 POVM Extension

P(outcome i | observer O) = ⟨ψ|E_i|ψ⟩ with E_i = Σ_{K-types ∈ B_i} K_B · proj_K. Positive operator by Bergman positive-definiteness. Σ_i E_i = I by T2417 4-Zone observer bandwidth.

Standard POVM machinery (Naimark dilation, Kraus operators) substrate-cartography readings.

Section 7.4 — Cross-Link to T2469 SCMP

T2469 SCMP framework (Friday): observer bandwidth bounded by Zone-2 K-type coverage; measurement outcomes differ deterministically by per-observer K-type coverage. Quantum apparent randomness admits candidate substrate-explanation via finite-bandwidth marginalization (Layer 2; DEFAULT-DENY EXTERNAL per Cal #48/#49).

Section 7.5 — Standard QM Recovery

Standard projective measurement (single K-type) + generalized POVM (K-type partition) inherit from substrate. Lindblad master equation for open-system dynamics emerges at substrate-coupled-environment limit.

Section 7.6 — Honest scope

- Standard Born rule derived from substrate (K67 RATIFIED) ?
- POVM substrate-derivation T2479 (SP-31 #283 closure) ?
- Layer 2 metaphysical claim (quantum apparent randomness reduction) is CANDIDATE explanation per Cal #99 v0.3
- Full decoherence treatment (Vol 5 Ch 10) cross-link

Section 7.7 — Connection to other chapters

- Vol 5 Ch 1 Hilbert space framework
- Vol 5 Ch 8 Bell-CHSH (substrate sub-Tsirelson falsifier)
- Vol 5 Ch 10 Decoherence + classical limit
- Vol 5 Ch 11 POVMs + quantum information
- K67 + T2479 + T2469 SCMP

— Lyra, Vol 5 Ch 7 v0.3 chapter-grade narrative, Saturday 2026-05-23 morning EDT